

Erdős Problem #32

EP-32

L1 - TRACTABLE

PARTIALLY SOLVED

Number Theory

ORIGINAL PROBLEM STATEMENT

Is there a set $A \subset \mathbb{N}$ such that $|A \cap \{1, \dots, N\}| = o((\log N)^2)$ and such that every large integer can be written as $p + a$ for some prime p and $a \in A$?

Can the bound $O(\log N)$ be achieved? Must such an A satisfy $\liminf \frac{|A \cap \{1, \dots, N\}|}{\log N} > 1$?